

NeuroDock

AI-Driven Molecular Docking Pipeline

MDD & Alzheimer's Disease Drug Discovery

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Pipeline: AutoDock Vina v1.2.5 | RDKit 2026 | Open Babel 3.1

Targets screened	1
Compounds screened	29
Docking runs	29
Docking hits (threshold)	23
Lead candidates (docking + ADMET)	6
Binding energy threshold	≤ -7.0 kcal/mol

Lead Candidates

The following compounds passed both the molecular docking energy threshold (≤ -7.0 kcal/mol) and all CNS ADMET filters including BBB permeability, TPSA, Lipinski Rule of Five, and PAINS screening.

■ Donepezil vs MAO-B

Property	Value	Threshold	Status
Binding Affinity (ΔG)	-8.84 kcal/mol	≤ -7.0	PASS
logBB (BBB permeability)	0.287	> -1.0	PASS
TPSA	38.77 Å ²	< 90 Å ²	PASS
MW	393.53 Da	≤ 450 Da	PASS
logP	4.75	-0.5 to 5.0	PASS
H-bond donors	0	≤ 3	PASS
H-bond acceptors	4	≤ 7	PASS
Rotatable bonds	6	≤ 8	PASS
SA Score	2.71	≤ 6.0	PASS
Lipinski violations	0	≤ 1	PASS
PAINS alerts	0	0	PASS

■ Chrysin vs MAO-B

Property	Value	Threshold	Status
Binding Affinity (ΔG)	-8.48 kcal/mol	≤ -7.0	PASS
logBB (BBB permeability)	-0.47	> -1.0	PASS
TPSA	70.67 Å ²	< 90 Å ²	PASS
MW	254.24 Da	≤ 450 Da	PASS
logP	2.87	-0.5 to 5.0	PASS
H-bond donors	2	≤ 3	PASS
H-bond acceptors	4	≤ 7	PASS
Rotatable bonds	1	≤ 8	PASS
SA Score	2.1	≤ 6.0	PASS
Lipinski violations	0	≤ 1	PASS
PAINS alerts	0	0	PASS

■ Resveratrol vs MAO-B

Property	Value	Threshold	Status
Binding Affinity (ΔG)	-8.35 kcal/mol	≤ -7.0	PASS
logBB (BBB permeability)	-0.307	> -1.0	PASS
TPSA	60.69 Å ²	< 90 Å ²	PASS
MW	228.25 Da	≤ 450 Da	PASS

logP	2.97	-0.5 to 5.0	PASS
H-bond donors	3	≤ 3	PASS
H-bond acceptors	3	≤ 7	PASS
Rotatable bonds	2	≤ 8	PASS
SA Score	2.11	≤ 6.0	PASS
Lipinski violations	0	≤ 1	PASS
PAINS alerts	0	0	PASS

■ Safinamide vs MAO-B

Property	Value	Threshold	Status
Binding Affinity (ΔG)	-8.13 kcal/mol	≤ -7.0	PASS
logBB (BBB permeability)	-0.453	> -1.0	PASS
TPSA	64.35 Å ²	< 90 Å ²	PASS
MW	302.35 Da	≤ 450 Da	PASS
logP	2.37	-0.5 to 5.0	PASS
H-bond donors	2	≤ 3	PASS
H-bond acceptors	3	≤ 7	PASS
Rotatable bonds	7	≤ 8	PASS
SA Score	2.27	≤ 6.0	PASS
Lipinski violations	0	≤ 1	PASS
PAINS alerts	0	0	PASS

■ Galantamine vs MAO-B

Property	Value	Threshold	Status
Binding Affinity (ΔG)	-7.74 kcal/mol	≤ -7.0	PASS
logBB (BBB permeability)	-0.2	> -1.0	PASS
TPSA	41.93 Å ²	< 90 Å ²	PASS
MW	287.36 Da	≤ 450 Da	PASS
logP	1.85	-0.5 to 5.0	PASS
H-bond donors	1	≤ 3	PASS
H-bond acceptors	4	≤ 7	PASS
Rotatable bonds	1	≤ 8	PASS
SA Score	4.23	≤ 6.0	PASS
Lipinski violations	0	≤ 1	PASS
PAINS alerts	0	0	PASS

■ Rivastigmine vs MAO-B

Property	Value	Threshold	Status
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Binding Affinity (ΔG)	-7.12 kcal/mol	≤ -7.0	PASS
logBB (BBB permeability)	0.073	> -1.0	PASS
TPSA	32.78 Å ²	< 90 Å ²	PASS
MW	250.34 Da	≤ 450 Da	PASS
logP	2.76	-0.5 to 5.0	PASS
H-bond donors	0	≤ 3	PASS
H-bond acceptors	3	≤ 7	PASS
Rotatable bonds	4	≤ 8	PASS
SA Score	2.67	≤ 6.0	PASS
Lipinski violations	0	≤ 1	PASS
PAINS alerts	0	0	PASS

Docking Results — All Compounds

All 29 ligand-target pairs ranked by binding affinity. Compounds with $\Delta G \leq -7.0$ kcal/mol are highlighted.

Rank	Ligand	Target	ΔG (kcal/mol)	RMSD l.b.	Docking
1	Hesperetin	MAO-B	-9.020	0.00	PASS
2	Hesperidin	MAO-B	-8.975	0.00	PASS
3	Rutin	MAO-B	-8.920	0.00	PASS
4	Donepezil	MAO-B	-8.844	0.00	PASS
5	Luteolin	MAO-B	-8.793	0.00	PASS
6	Quercetin	MAO-B	-8.692	0.00	PASS
7	Fisetin	MAO-B	-8.654	0.00	PASS
8	Myricetin	MAO-B	-8.606	0.00	PASS
9	Baicalein	MAO-B	-8.591	0.00	PASS
10	Genistein	MAO-B	-8.541	0.00	PASS
11	Kaempferol	MAO-B	-8.537	0.00	PASS
12	Chrysin	MAO-B	-8.484	0.00	PASS
13	Apigenin	MAO-B	-8.471	0.00	PASS
14	Catechin	MAO-B	-8.418	0.00	PASS
15	Epicatechin	MAO-B	-8.366	0.00	PASS
16	Resveratrol	MAO-B	-8.352	0.00	PASS
17	7,8-DHF	MAO-B	-8.315	0.00	PASS
18	7,8-Dihydroxyflavone	MAO-B	-8.281	0.00	PASS
19	Curcumin	MAO-B	-8.199	0.00	PASS
20	Safinamide	MAO-B	-8.134	0.00	PASS
21	Epigallocatechin gallate	MAO-B	-8.036	0.00	PASS
22	Galantamine	MAO-B	-7.744	0.00	PASS
23	Rivastigmine	MAO-B	-7.124	0.00	PASS
24	Naringenin	MAO-B	-6.905	0.00	FAIL
25	NP-alpha1	MAO-B	-6.755	0.00	FAIL
26	Rasagiline	MAO-B	-6.618	0.00	FAIL
27	Caffeic acid	MAO-B	-6.357	0.00	FAIL
28	Ferulic acid	MAO-B	-6.263	0.00	FAIL
29	Selegiline	MAO-B	-5.787	0.00	FAIL

ADMET & CNS Drug-Likeness Analysis

ADMET properties calculated per compound using RDKit. logBB estimated via Clark (1999) empirical model: $\log BB = -0.0148 \times TPSA + 0.152 \times \log P + 0.139$. PAINS alerts detected using the Baell & Holloway (2010) filter set.

Compound	MW	logP	HBD	HBA	TPSA	logBB	SA	PAINS	ADMET
Hesperetin	302.28	2.52	3	6	96.22 Å ²	-0.902	2.91	0	FAIL
Hesperidin	610.57	-1.16	8	15	234.29 Å ²	-3.504	4.82	0	FAIL
Rutin	610.52	-1.69	10	16	269.43 Å ²	-4.105	4.78	1	FAIL
Donepezil	393.53	4.75	0	4	38.77 Å ²	0.287	2.71	0	PASS
Luteolin	286.24	2.28	4	6	111.13 Å ²	-1.159	2.42	1	FAIL
Quercetin	302.24	1.99	5	7	131.36 Å ²	-1.503	2.54	1	FAIL
Fisetin	286.24	2.28	4	6	111.13 Å ²	-1.159	2.36	1	FAIL
Myricetin	318.24	1.69	6	8	151.59 Å ²	-1.847	2.73	1	FAIL
Baicalein	270.24	2.58	3	5	90.9 Å ²	-0.815	2.25	1	FAIL
Genistein	270.24	2.58	3	5	90.9 Å ²	-0.815	2.3	0	FAIL
Kaempferol	286.24	2.28	4	6	111.13 Å ²	-1.159	2.37	0	FAIL
Chrysin	254.24	2.87	2	4	70.67 Å ²	-0.47	2.1	0	PASS
Apigenin	270.24	2.58	3	5	90.9 Å ²	-0.815	2.25	0	FAIL
Catechin	290.27	1.55	5	6	110.38 Å ²	-1.26	3.34	1	FAIL
Epicatechin	290.27	1.55	5	6	110.38 Å ²	-1.26	3.34	1	FAIL
Resveratrol	228.25	2.97	3	3	60.69 Å ²	-0.307	2.11	0	PASS
7,8-DHF	254.24	2.87	2	4	70.67 Å ²	-0.47	2.05	1	FAIL
7,8-Dihydroxyflavone	254.24	2.87	2	4	70.67 Å ²	-0.47	2.1	1	FAIL
Curcumin	368.39	3.37	2	6	93.06 Å ²	-0.726	2.43	0	FAIL
Safinamide	302.35	2.37	2	3	64.35 Å ²	-0.453	2.27	0	PASS
Epigallocatechin gallate	458.38	2.23	8	11	197.37 Å ²	-2.443	3.74	1	FAIL
Galantamine	287.36	1.85	1	4	41.93 Å ²	-0.2	4.23	0	PASS
Rivastigmine	250.34	2.76	0	3	32.78 Å ²	0.073	2.67	0	PASS
Naringenin	272.26	2.51	3	5	86.99 Å ²	-0.767	2.83	0	PASS
NP-alpha1	220.32	1.41	1	3	24.5 Å ²	-0.009	1.86	0	PASS
Rasagiline	171.24	1.9	1	1	12.03 Å ²	0.249	2.77	0	PASS
Caffeic acid	180.16	1.2	3	3	77.76 Å ²	-0.83	2.04	1	FAIL
Ferulic acid	194.19	1.5	2	3	66.76 Å ²	-0.621	1.87	0	PASS
Selegiline	159.23	1.75	0	1	3.24 Å ²	0.357	1.97	0	PASS

Failure Details

Hesperetin: TPSA=96.22 Å² (need <90 for CNS)

Hesperidin: MW=610.57 (need ≤450 Da for CNS); logP=-1.16 (need -0.5–5.0); HBD=8 (need ≤3 for CNS); HBA=15 (need ≤7); TPSA=234.29 Å² (need <90 for CNS); logBB=-3.504 (need >-1.0 for CNS); Lipinski violations=3 (need ≤1)

Rutin: MW=610.52 (need ≤ 450 Da for CNS); logP=-1.69 (need -0.5–5.0); HBD=10 (need ≤ 3 for CNS); HBA=16 (need ≤ 7); TPSA=269.43 Å² (need <90 for CNS); logBB=-4.105 (need >-1.0 for CNS); Lipinski violations=3 (need ≤ 1); PAINS alerts=1: catechol_A(92)

Luteolin: HBD=4 (need ≤ 3 for CNS); TPSA=111.13 Å² (need <90 for CNS); logBB=-1.159 (need >-1.0 for CNS); PAINS alerts=1: catechol_A(92)

Quercetin: HBD=5 (need ≤ 3 for CNS); TPSA=131.36 Å² (need <90 for CNS); logBB=-1.503 (need >-1.0 for CNS); PAINS alerts=1: catechol_A(92)

Fisetin: HBD=4 (need ≤ 3 for CNS); TPSA=111.13 Å² (need <90 for CNS); logBB=-1.159 (need >-1.0 for CNS); PAINS alerts=1: catechol_A(92)

Myricetin: HBD=6 (need ≤ 3 for CNS); HBA=8 (need ≤ 7); TPSA=151.59 Å² (need <90 for CNS); logBB=-1.847 (need >-1.0 for CNS); PAINS alerts=1: catechol_A(92)

Baicalein: TPSA=90.9 Å² (need <90 for CNS); PAINS alerts=1: catechol_A(92)

Genistein: TPSA=90.9 Å² (need <90 for CNS)

Kaempferol: HBD=4 (need ≤ 3 for CNS); TPSA=111.13 Å² (need <90 for CNS); logBB=-1.159 (need >-1.0 for CNS)

Apigenin: TPSA=90.9 Å² (need <90 for CNS)

Catechin: HBD=5 (need ≤ 3 for CNS); TPSA=110.38 Å² (need <90 for CNS); logBB=-1.26 (need >-1.0 for CNS); PAINS alerts=1: catechol_A(92)

Epicatechin: HBD=5 (need ≤ 3 for CNS); TPSA=110.38 Å² (need <90 for CNS); logBB=-1.26 (need >-1.0 for CNS); PAINS alerts=1: catechol_A(92)

7,8-DHF: PAINS alerts=1: catechol_A(92)

7,8-Dihydroxyflavone: PAINS alerts=1: catechol_A(92)

Curcumin: TPSA=93.06 Å² (need <90 for CNS)

Epigallocatechin gallate: MW=458.38 (need ≤ 450 Da for CNS); HBD=8 (need ≤ 3 for CNS); HBA=11 (need ≤ 7); TPSA=197.37 Å² (need <90 for CNS); logBB=-2.443 (need >-1.0 for CNS); Lipinski violations=2 (need ≤ 1); PAINS alerts=1: catechol_A(92)

Caffeic acid: PAINS alerts=1: catechol_A(92)

Methods & Pipeline Overview

Protein Preparation

Protein structures were downloaded from the RCSB Protein Data Bank. Crystallographic water molecules and co-crystallised ligands were removed. Structures were converted to PDBQT format using Open Babel 3.1 with Gasteiger partial charge assignment.

Ligand Preparation

Ligands were prepared from SMILES notation using RDKit 2026. 3D conformers were generated using the ETKDGv3 algorithm with MMFF94 force field geometry optimisation. Conversion to PDBQT format was performed with Open Babel.

Molecular Docking

Molecular docking was performed using AutoDock Vina v1.2.5. Exhaustiveness was set to 16 with 9 binding modes per run. Compounds with binding affinity ≤ -7.0 kcal/mol were considered hits.

ADMET Filtering

ADMET properties were calculated using RDKit descriptors. Blood-brain barrier permeability was estimated using the Clark (1999) logBB model. PAINS (Pan-Assay Interference Compounds) screening used the Baell & Holloway (2010) filter catalogue. CNS-specific thresholds were applied: MW ≤ 450 Da, TPSA < 90 Å², logBB > -1.0 , HBD ≤ 3 , rotatable bonds ≤ 8 .

Targets

PDB ID	Target	Disease Relevance
2V61	MAO-B	Oxidative neuroprotection — dopamine catabolism

Software & Tools

AutoDock Vina 1.2.5 | Open Babel 3.1 | RDKit 2026 | Python 3.12 | ReportLab (PDF generation)